OSCAR OLIVA

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EDUCATION

University of Pittsburgh – Swanson School of Engineering April 2025

- Bachelor of Science in Mechanical Engineering
- Minor in Bioengineering

PROJECTS / ACCOMPLISHMENTS

- ZOLL: Conductively Coated Fabric Electrode – Patent Pending | Co-Inventor
- Engineering Portfolio is available via the QR code at the top of this document or at: https://oaoliva.github.io/

ENGINEERING EXPERIENCE

Wood

Field Vibration Analyst *May 2025 - Present*

- Visited 10+ client sites to collect vibration data on gas compressors, focusing on piping systems and small bores piping.
- Analyzed vibration data using FFT and compared the spectrum to the systems MNF and operating deflection shape.
- Developed targeted solutions for identified problem areas in a comprehensive report and presented findings to stakeholders.
- Successfully completed engineering projects on time, adhering to the proposed budgets, and meeting all project milestones.

Human Movement and Balance Laboratory, Department of Bioengineering **Pittsburgh, PA**

Research Assistant (only while enrolled in classes) *November 2021 – April 2025*

- Integrated a Raspberry Pi and camera into a portable scanner which used frustrated total internal reflection (FTIR) technology, enabling automated capture, storage, and analysis of tread wear images upon user interaction.
- Developed MATLAB-based image processing software to identify worn shoe regions from FTIR data and utilized Python to correlate image brightness with pressure exerted on the shoe.
- Adapted tribology testing equipment to evaluate friction on various ladder rungs under dry and contaminated conditions; designed custom supports and established a robust testing protocol.
- Analyzed tribology data in MATLAB to quantify relationships between surface patterns, materials, rung geometries, and coefficients of friction.
- Conducted gait analysis using Vicon motion capture to study participant movement during ladder ascent, descent, and unexpected slip scenarios.

ZOLL LifeVest **Pittsburgh, PA**

Human Factors Engineering CO-OP (three 4-month rotations) *September 2022 – April 2024*

- Conducted post-market surveillance analysis on an existing product, contributing to the HFE/UE validation report submitted to the FDA for the next-generation product.
- Developed protocols for wear test evaluations and moderated studies by interviewing participants to assess comfort, usability, and the effectiveness of error-prevention features of the LifeVest system.
- Leveraged R and Power BI to analyze study data, prepared detailed reports, presented findings, and collaborated with a multidisciplinary engineering team to address critical use errors.

Advanced Research / Advanced Development *May 2024 – July 2024*

- Designed a prototype LifeVest garment to enhance patient comfort while maintaining ECG quality by incorporating fabric electrodes and relocating circuitry to the garment's exterior.
- Applied 3D scanning to capture patient body measurements for garment sizing and analyzed correlations between ECG wave amplitudes and electrode placement across the body.

Desapro Inc. **Rockledge, FL**

Mechanical Engineering Intern May 2022 – August 2022

- Drafted SolidWorks models of aluminum transit cases and prepared detailed production-ready drawings.
- Analyzed sheet metal-fabricated cases for manufacturability and implemented design adjustments using SolidWorks, resulting in improved production efficiency.
- Collaborated with customers to deliver custom cases tailored to specific size, weight, and load-bearing requirements, leading both design and manufacturing efforts.
- Designed a temperature-controlled case for drone transport of blood bags to soldiers in combat scenarios.
- Designed and managed the manufacturing of a vibration-dampening case using advanced materials to reduce impact shock in sensitive equipment

SKILLS

● SolidWorks ● Fusion360 ● MATLAB ● Python ● R ● Microsoft Office ●Instron Testing
● Rapid Prototyping – Soldering, Sewing, Injection molding ● Technical Writing/Presenting ● Spanish (Fluent)